Leakage-Controlled Multi-Action Three-Tower Retrieval on KuaiRand-Pure

Summary. On KuaiRand-Pure we retrieve videos and their background-music IDs for two actions, long view (LV) and like, giving four full-catalog tasks. A three-tower model (user / video / music) is compared with Popularity and implicit ALS. All models are scored by one shared evaluator on identical test users, with 95% confidence intervals (CIs) from a 1,000-resample paired user bootstrap. Findings for the main model **U0** (user ID + strict-past history, no user metadata), on the formal **test** split:

- It beats Popularity on all four tasks.
- Against Pure ALS it is **worse on both LV tasks** (Recall@20 0.1124 vs 0.1244 on video LV, a −0.012 difference, significant) and better on both like tasks.
- On like it is **not significantly different** from Aux ALS, an ALS that adds the LV signal. This does not mean the two are equivalent.
- In a controlled cold-video holdout, content features rank new videos well *among themselves* but are **not significantly better than random on the full catalog**.
- Removing history hurts LV; removing the music input to the video tower **improves** LV.

Post-hoc exploratory findings (validation only, three training seeds per setting, §2.4). The study was designed after the test results were seen, and it does not change any test number:

- **Removing the music-to-video input consistently improved long-view validation Recall@20 across three seeds, without systematically harming like retrieval** (+0.0068 video LV, +0.0072 music LV). Its long-view gap to Pure ALS is no longer significant, which is not a proof of equivalence. The mechanism is not identified.
- Training-seed std (about 0.004 in the 4-task mean) is as large as most tuning effects. lr 5e-4 gains +0.0048, mostly on like. $\tau = 0.1$ trades like for LV.

Unabridged report and audit evidence: `REPORT_FULL.md`. All metrics and differences: `tables/`.

1. Design

1.1 Data and cleaning

Inputs. The loader reads only the two standard logs, `user_features_pure.csv` and `video_features_basic_pure.csv`. It refuses the random log and the monthly statistics file (§1.2).

Data. There are 1,436,609 raw rows from 27,077 users on 7,551 videos, and the catalog has 7,583 videos. Long view is positive on 33.2% of rows and like on only 1.85%.

Music. It is almost one-to-one with video: of 7,201 non-zero music IDs, only 112 are shared, by at most 15 videos. `music_id = 0` (204 videos) is treated as a "no music" sentinel. This is our assumption; the data dictionary does not confirm it. It is never a music candidate or music ground truth, and gets a learnable NO_MUSIC embedding.

Cleaning. Before the split, keyed on (user, video, time):

- 21,987 exact duplicates are collapsed;
- 1,486 conflict groups each keep one whole representative row;
- 2 label-conflicting events are dropped and counted;
- there is no k-core filtering.

This leaves 1,413,134 events, with all users and videos kept.

1.2 Temporal split and leakage control

Split. The split is global and chronological, at CST midnights; $t < T$ goes to the earlier split. No timestamp spans two splits.

split	days (2022)	events	share	LV / like positives
train	04-09 – 04-16	997,078	70.6%	339,386 / 18,771
val	04-17 – 04-25	208,393	14.7%	69,354 / 3,980
test	04-26 – 05-08	207,663	14.7%	65,812 / 3,610

No future-leaking aggregates. `video_features_statistic_pure.csv` aggregates each video's plays, likes and shares over the whole log month. Using it at time t would encode test-period behaviour, so the loader refuses it, and none of its 51 columns appears in any processed file. The video tower reads only static metadata (type, upload type/date, music type, tags, duration, size) and IDs.

Train-only fitting.

- Fitted on train events only (assertion-guarded): every vocabulary, scaler, Popularity count and ALS factor.
- Validation only selects the ALS grid point and the three-tower epoch.
- Test is evaluated once, after selection, and every run records `test_used_for_selection = false`.

Strict-past history and seen mask.

- One function builds histories at the event time (training), at T_val (train events) and at T_test (train + val events).
- It keeps only t < t_pred, so same-timestamp siblings are also excluded (10% of rows share one), and it caps the input at the latest 100 events.
- Feedback younger than 20 min is hidden, because the dataset does not say when it becomes known (an assumption).
- The seen mask is built separately from *all* earlier interactions, uncapped.

User metadata (U0 vs U1). `user_features_pure.csv` is one undated snapshot. One user's registration age bounds it at no later than 2022-05-02, inside the test period; there is no lower bound. Of the 21 candidate fields:

- 2 are time-dependent by definition (`register_days_range`, cumulative `is_video_author`);
- 19 have unknown historical availability;
- none can be shown to be available at prediction time.

This is not evidence that every field leaks. We report **U0** (main: user ID + history) and **U1** (U0 + the 21 fields, same configuration and seed, with a leakage caveat).

1.3 Three-tower model

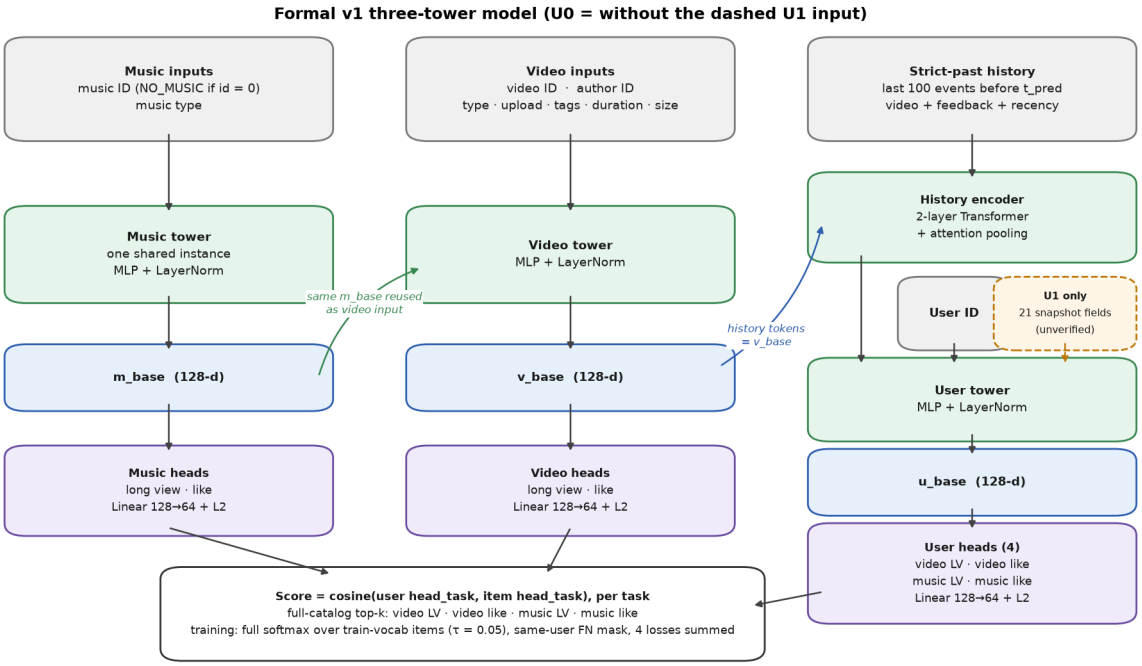


Figure 1. The formal three-tower architecture (U0; U1 adds the snapshot fields to the user tower).

Towers.

- **Music tower:** music ID + type → MLP + LayerNorm → 128-d.
- **Video tower:** video ID, author, metadata and the music-tower output → 128-d. There is **one shared music tower**: it runs once per catalog forward, and each video reuses that output, so video losses also train it.
- **User tower:** user ID + a 2-layer Transformer with attention pooling over the last 100 history events (each token is the shared video embedding plus feedback, play ratio, recency and position) → 128-d. U1 adds 21 field embeddings.

Heads. Each target × action has its own `Linear(128→64) + L2` head (user 4, video 2, music 2), and the score is cosine similarity.

Unseen IDs. The embedding rows for padding and out-of-vocabulary (OOV, not seen in training) IDs are fixed at zero, and ID dropout 0.1 trains this path, so unseen videos are scored from content. The model has 3.77M parameters.

1.4 Contrastive objective: duplicates, false negatives, task balance

Samples. Every train event positive for either action is a sample (345,365; 18,771 likes), with four task masks (music tasks need `music_id ≠ 0`) and its own rolling strict-past history.

Loss. Per task, an exact full softmax over all train-vocabulary items (7,529 videos, 7,151 music IDs), so no sampling or logQ correction is needed:

$$\ell_r = -\log \frac{\exp(\cos(u_r, i_{y_r})/\tau)}{\sum_{c \notin \mathrm{FN}(r)} \exp(\cos(u_r, i_c)/\tau)}, \quad \tau = 0.05, \qquad \mathrm{FN}(r) = \mathrm{Pos}_{\mathrm{train}}(u_r, \mathrm{task}) \setminus \{y_r\}$$

Duplicate positives. Candidate columns are unique items, so rows with the same label share one column. This covers repeats by one user, different users, and videos sharing a music ID. A row's own label is never masked.

False negatives.

- `Pos_train(u, task)` holds u's *train* positives for the same target and action. Those columns (except the label) are set to –inf, so they get zero weight and zero gradient.
- Validation and test positives are never used.
- Pos_train includes positives that occur **after** the sample's time (about 50% of masked columns). It is a *supervision* mask that stops the loss from pushing a user's other known positives down as negatives. It is **never a model input**: history stays strictly in the past.
- On four real batches, an independent recomputation matched every label and mask, and the loss to 8e-7.

Task balance.

- The loss is the equal-weight sum of the four per-task *mean* losses; like rows are not oversampled.
- A batch has about 1,006 LV rows but 56 like rows, so each like row carries about 18× the weight of an LV row: the actions are balanced at the task level.
- Like also benefits from towers and history shared with LV; only the heads are action-specific.
- The 1:1 weight was not tuned, and like gradients are noisier.

Optimisation. AdamW (lr 1e-3), batch 1,024, FP32, deterministic, at most 30 epochs with patience 3 on mean validation R@20, one seed. **v1 was not tuned**, although its validation LV R@20 was below ALS (video 0.1260 vs 0.1339, music 0.1308 vs 0.1361), which the design allows as a trigger for targeted adjustments. These were explored afterwards on a separate branch (§2.4).

1.5 Evaluation protocol

Ground truth. Each user is represented once at T_test. The ground truth GT(u) is the de-duplicated set of u's test positives, minus three kinds of item:

- out-of-catalog (OOC) items, i.e. items outside the ranked catalog;
- the music sentinel;
- items u interacted with before T_test (the *seen mask*, which is also applied at ranking).

Catalog and metrics. All 7,583 videos or 7,201 music IDs are ranked, and train-unseen items stay in the catalog and GT. Recall@K (R@K) = hits in the top K / |GT|. NDCG@K uses binary gain with IDCG over min(|GT|, K), so it also rewards ranking hits higher. Users are weighted equally, and ties break by catalog row. A *difference* is model A – model B on the same users (an absolute difference in the metric). It is *significant* if its 95% CI excludes 0; a non-significant difference does not imply equivalence.

Population. Evaluated users have non-empty GT and at least one history event; users with no history are reported separately.

task	users	GT pairs	seen-excluded GT pairs	OOC positive events	sentinel-excluded events	cold users (separate)
video LV	18,752	63,281	859	0	–	437

task	users	GT pairs	seen-excluded GT pairs	OOC positive events	sentinel-excluded events	cold users (separate)
video like	1,924	3,380	63	0	–	68
music LV	18,595	61,595	913	0	1,670	434
music like	1,895	3,307	72	0	67	67

Fair comparison.

- Every model on a task must match one `frame_id`, a hash of users, GT and mask.
- **Popularity** uses train counts.
- **Pure ALS** uses train-only item factors for the task's own action. User vectors are folded in from pre-T_test history, and the same 8-point validation grid is used on every task. Users or items without signal score –inf, not random vectors.
- **Aux ALS** (like tasks only) adds 0.3 × the LV signal to the like signal.
- CIs come from 1,000 user resamples shared (paired) by all models on a frame.

2. Results

§2.1–2.3 report formal **test** results from `results/m5/`. §2.4 is a separate post-hoc **validation** study. In difference tables, a star (*) marks a 95% CI that excludes 0 (significant). Full metrics are in `tables/`.

2.1 Main comparison

Table 1. Absolute test R@20 / NDCG@20 per model. Bold marks the best value per column; "–" means Aux ALS is defined for like only.

model	video LV	video like	music LV	music like
Pop	0.0806 / 0.0457	0.0716 / 0.0403	0.0856 / 0.0479	0.0726 / 0.0412
Pure ALS	**0.1244** / 0.0686	0.0555 / 0.0282	**0.1282** / 0.0705	0.0586 / 0.0303
Aux ALS	–	**0.1008 / 0.0469**	–	**0.1054** / 0.0463
U0 (main)	0.1124 / 0.0676	0.0913 / 0.0417	0.1162 / 0.0691	0.1022 / **0.0472**
U1 (+ snapshot)	0.1274 / 0.0744	0.0859 / 0.0397	0.1321 / 0.0768	0.0864 / 0.0407

Table 2. Paired test differences in R@20, A – B [95% CI]. These are absolute differences of the values in Table 1, on the same users.

comparison (A – B)	video LV	video like	music LV	music like
U0 – Pop	+0.032* [+0.029, +0.035]	+0.020* [+0.007, +0.032]	+0.031* [+0.027, +0.034]	+0.030* [+0.015, +0.043]
U0 – Pure ALS	**–0.012*** [–0.016, –0.008]	+0.036* [+0.022, +0.049]	**–0.012*** [–0.016, –0.008]	+0.044* [+0.029, +0.058]
U0 – Aux ALS	–	–0.009 [–0.025, +0.005]	–	–0.003 [–0.019, +0.014]
U1 – Pure ALS	+0.003 [–0.001, +0.007]	+0.030* [+0.017, +0.043]	+0.004 [–0.000, +0.008]	+0.028* [+0.015, +0.042]
U1 – U0	+0.015* [+0.012, +0.018]	–0.005 [–0.016, +0.005]	+0.016* [+0.013, +0.019]	–0.016* [–0.027, –0.004]

U0 vs Pure ALS. On LV, U0's NDCG@20 difference is not significant (–0.001 / –0.001), so U0 trails ALS in recall depth rather than in ranking its hits.

Effect of history length. The deficit is a heavy-user effect: U0 – Pure ALS on LV is +0.028* / +0.037* for users with 1–5 history events, but –0.017* / –0.016* for users with more than 20, who are 72% of LV users.

U1. It reaches Pure-ALS level on LV, but this may partly be leakage (§1.2). **We do not claim that the three-tower model beats ALS.**

2.2 Controlled cold-video experiment

Setup.

- H is 750 train-seen videos (10%), drawn from train data only and stratified by train interaction count.
- Every H interaction is removed from targets, all histories (including at test time), candidates, false-negative sets, vocabularies and the cold Pop/ALS fits; 12 automated checks verify the removal.
- The cold model selects its epoch on validation without H ground truth.
- GT_H is each user's test positives on H that the user had not seen before T_test (5,936 LV / 305 like users). It is ranked against (a) the full catalog and (b) H only.

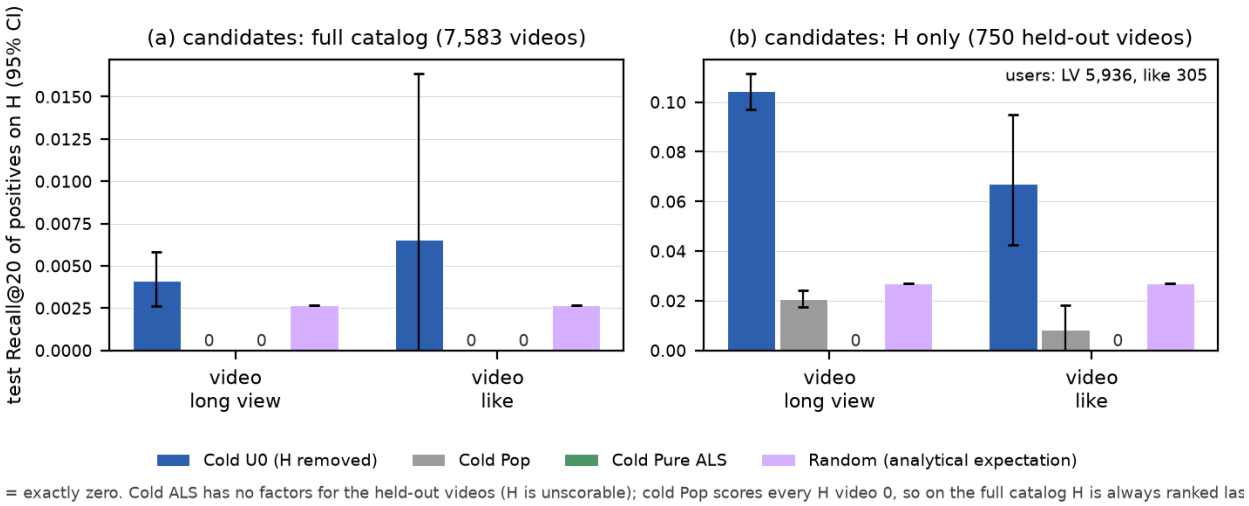


Figure 2. Test Recall@20 on GT_H with 95% CIs. Left: candidates are the full catalog; right: candidates are the 750 H videos only. All models shown were trained without H. Cold Pure ALS is 0 by construction, because it has no factors for H.

Table 3. Cold-video test R@20: absolute values for the cold U0 and two cold-condition baselines, then the paired differences cold U0 – baseline [95% CI]. "Random" is the closed-form per-user expectation of a uniformly random ranking.

task	candidates	cold U0	cold Pop	random	cold U0 – cold Pop	cold U0 – random
video LV	full catalog	0.0041	0.0000	0.0027	+0.0041* [+0.0026, +0.0058]	+0.0014 [−0.0000, +0.0032]
video LV	H only	0.1045	0.0207	0.0269	+0.0838* [+0.0756, +0.0923]	+0.0776* [+0.0703, +0.0848]
video like	full catalog	0.0066	0.0000	0.0027	+0.0066 [+0.0000, +0.0164]	+0.0039 [−0.0027, +0.0137]
video like	H only	0.0672	0.0082	0.0268	+0.0590* [+0.0328, +0.0869]	+0.0404* [+0.0158, +0.0682]

H only. Metadata, author and music carry a real ranking signal among new videos. Here cold Pop is a fixed catalog order, not a random ranking, since all its scores are 0.

Full catalog. The cold model is not significantly different from random, because cold videos rarely reach the top 20. Cold ALS cannot score H at all.

Context. The normally trained U0 reaches 0.111 / 0.433 on the same frame, but it saw H in training, so it is context only.

2.3 Ablations

Each ablation changes one configuration flag; everything else is as in U0. The first row gives U0's absolute test values (from Table 1). The other rows give the paired test difference *ablation − U0*; the CIs are in `tables/t2_paired_diffs.md`.

	video LV	video like	music LV	music like
U0, absolute R@20 / NDCG@20	0.1124 / 0.0676	0.0913 / 0.0417	0.1162 / 0.0691	0.1022 / 0.0472
no history − U0, R@20	−0.012*	−0.002	−0.011*	−0.017*
no history − U0, NDCG@20	−0.010*	+0.001	−0.009*	−0.005
no music input − U0, R@20	**+0.010***	+0.007	**+0.009***	−0.008
no music input − U0, NDCG@20	**+0.004***	+0.001	**+0.004***	−0.004

History is the main source of LV personalisation.

Music input. Removing it from the video tower **improves** LV significantly, a negative result for that design choice. We report it as is and kept the specified architecture. §2.4 shows that the effect replicates across seeds on validation.

2.4 Post-hoc exploratory study (validation only)

Scope.

- After the test results above had been analysed, U0 was retrained on a separate branch under single-setting changes and seed replicates. This gives 11 runs, all with exit code 0 and bitwise export/direct parity.
- **All numbers below are validation metrics**, on the same validation users and masks as the baselines. They are read by scripts that refuse test files. They are not test results and do not replace U0.
- Two kinds of uncertainty are reported separately. *mean ± std* is over three training seeds; it describes training-seed variability and is descriptive only (n = 3). CIs (and *) are paired user bootstraps; they cover user sampling, not training randomness.
- Same-config reruns reproduce the formal U0 (job 27) and the formal no-music run (job 35) bitwise.

Table 4. Validation Recall@20 per setting. Absolute rows: mean ± std over three training seeds. Δ rows: the mean same-seed difference vs U0, with the min–max over the three seeds in braces (a seed range, not a CI). The τ row is a single seed, compared with U0 seed 0, and its * is the bootstrap CI.

setting	video LV	music LV	video like	music like	4-task mean
U0 (lr 1e-3, τ 0.05)	0.1256 ± 0.0014	0.1285 ± 0.0073	0.1147 ± 0.0069	0.1190 ± 0.0037	0.1220 ± 0.0036
no music input	**0.1324** ± 0.0018	**0.1357** ± 0.0005	0.1218 ± 0.0039	0.1246 ± 0.0066	**0.1286** ± 0.0018
Δ no-music − U0	+0.0068 {+0.004, +0.009}	+0.0072 {+0.001, +0.016}	+0.0071 {−0.005, +0.017}	+0.0056 {+0.003, +0.011}	+0.0066 {+0.003, +0.009}
lr 5e-4	0.1266 ± 0.0046	0.1320 ± 0.0068	0.1229 ± 0.0061	0.1253 ± 0.0076	0.1267 ± 0.0035
Δ lr 5e-4 − U0	+0.0010 {−0.005, +0.004}	+0.0035 {−0.010, +0.015}	+0.0082 {+0.003, +0.016}	+0.0063 {+0.002, +0.011}	+0.0048 {+0.003, +0.006}
τ 0.1 (seed 0), Δ vs U0 seed 0	+0.0085*	+0.0094*	−0.0114*	−0.0083	−0.0005

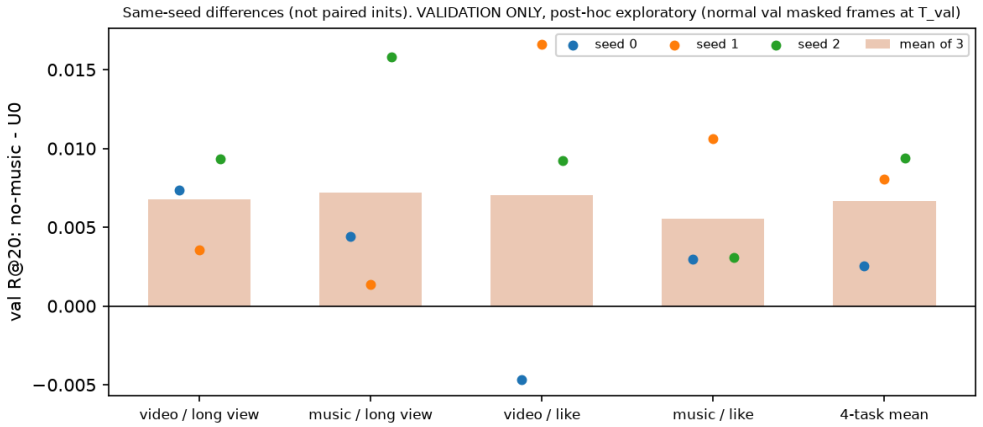


Figure 3. Validation Recall@20 of no-music minus U0 at the same seed. Dots are seeds 0/1/2; bars are the 3-seed mean. The architectures differ in layer size, so the same seed does not give the same initialisation.

No-music: the most consistent finding.

- Both long-view tasks and the 4-task mean are higher on 3/3 seeds. The no-music seed ranges lie entirely above the U0 ranges.
- Like is not systematically lower: 5/6 differences are positive, and the one negative has a CI that includes 0.
- The lead holds at nearly every epoch and at per-task best epochs, so it is not an artefact of early stopping on the 4-task mean.

Comparison with ALS on validation. We separate "numerically above" from "CI-supported".

- Every U0 seed is significantly below Pure ALS on video LV, and 2/3 are below on music LV.

- With no music input, all six long-view R@20 CIs vs Pure ALS include 0. The gap is no longer significant, but this is not a demonstration of equivalence, and no run has a CI-supported R@20 advantage.
- On long-view NDCG@20, no-music is above Pure ALS on 6/6 seed-task pairs (+0.003 to +0.004, nominal 95% CIs exclude 0). This is a secondary metric, uncorrected for multiple comparisons and not confirmed on test.

Tuning.

- Across the three U0 seeds, the 4-task mean spans 0.0072. This is the observed range, not a statistical threshold, but it exceeds user-bootstrap uncertainty, so single-seed gains of that size should be read with caution.
- lr 5e-4 is higher on 3/3 seeds, but by about one seed std, mostly on like, and LV changes sign across seeds.
- τ = 0.1 (one seed) moves performance from like to LV without changing the mean.

Limits.

- The study is post-hoc: it was designed after the test results, it was partly motivated by the test-side no-music ablation, and choosing the best setting carries a winner's-curse bias.
- The mechanism is not identified, because removing the music input also removes the video-loss gradient into the shared music tower (E5).

Details: `REPORT_FULL.md` §2.5 and `docs/EXPLORATORY_RESULTS.md`.

3. Error analysis

Post-hoc breakdowns of the test results (`results/final_analysis/ERROR_ANALYSIS.md`). They were not used for any choice in the formal v1, but they did motivate the later exploratory study (§2.4).

E1. The like gain over Pure ALS is mostly coverage.

- *Observed:* 608 of the 1,924 video-like users have no prior like, so Pure ALS cannot score them. On them, U0 − Pure ALS is +0.086*. On users with a prior like, U0 − Pure ALS is only +0.012 [−0.005, +0.029], and U0 is *below* Aux ALS (−0.022*).
- *Interpretation:* U0's like advantage is transfer from LV, which Aux ALS achieves explicitly.

E2. The LV deficit grows with history length.

- *Observed:* in the >20 bucket on video LV, U0 − Pure ALS is −0.016* when the history fits in 100 events and −0.021* when it was truncated. ALS fold-in uses every interaction.
- *Hypotheses (untested):* the capped, pooled representation loses long-range preference, and training histories (0–7 days) are shorter than test histories (16 days).

E3. The empty-history path is not popularity-calibrated.

- *Observed:* on the 437 video-LV users without history, U0 reaches R@20 0.039 versus Pop 0.142. Their shared embedding's scores correlate with popularity at only 0.48 (Spearman; no-history ablation: 0.92, R@20 0.147), and U0 sees that exact path only about 1,100 times per epoch.
- *Interpretation:* the fallback is under-trained.

E4. Cold videos are under-scored against warm ones.

- *Observed:* H is 9.9% of the catalog but gets 0.9% of the cold model's top-20 slots, and the best GT_H item has median rank 798. For 72% of H, all three IDs (video, author, music) are unknown, while training drops all three at once with probability 0.001.
- *Hypothesis:* ID-free embeddings are off-distribution.

E5. Music adds little beyond video here.

- *Observed:* per-user U0 R@20 on music LV and video LV correlates at 0.88.
- *Hypothesis (unverified):* with near one-to-one music, the music input acts as a competing second ID path (or its shared-tower gradients interfere). The no-music gain replicates on 3 validation seeds (§2.4), but these explanations are not yet separated.

4. Limitations

- **Main LV result is negative:** U0 is below Pure ALS on LV, and on like it is not significantly different from Aux ALS. v1 was not tuned.
- **User metadata:** U1's fields cannot be shown to be available at prediction time, so its LV gain may partly be leakage. U0 avoids the risk but uses no user metadata, a documented deviation from "ID + usable metadata + history".
- **Cold videos:** the positive result holds only for the H-only candidate set.
- **One seed per formal model:** test CIs reflect user sampling only. The exploratory three-seed study (§2.4) puts the training-seed std at about 0.004 (4-task mean validation R@20). It is post-hoc and validation-only, so no explore result is an independent test confirmation.
- **Assumptions:** a 20-minute feedback delay; the false-negative (FN) mask uses later train positives as supervision; ALS selection hit the grid edge (64 factors).
- **Numerics and scope:** re-scoring the Linux exports on macOS changes at most 2 users per frame (test means ≤ 1.1e-7). KuaiRand-Pure keeps candidate-pool videos only.

5. What I'd do next

1. **Music fusion:** the no-music gain replicates on 3 seeds (§2.4). Next, test a stop-gradient fusion to separate gradient interference from input competition and scale, then re-confirm on a fresh held-out period. U1 seed replicates are still to do.
2. **Empty-history fallback (E3):** a learned popularity bias or a Pop fallback for users without history.
3. **Heavy-user LV (E2):** per-epoch frozen training histories, a longer cap or a long-term mean-pooled summary, and an ensemble with ALS fold-in.
4. **Like (E1):** an LV→like auxiliary target or score mixing with Aux ALS, plus validation-tuned task weights and like oversampling.
5. **Cold calibration (E4):** whole-item ID dropout and a content-only auxiliary loss.
6. **Pretrained representations and stronger architectures (longer term):** pretrained content or sequence encoders, for example LLM or vision item embeddings for cold items, evaluated on the same frozen frames.

Reproducibility. U0/U1 were trained at commit `9218a76`, the cold and ablation runs at `671dde9`, and the exploratory runs on branch `explore/s9-opt` (`1051f86`, `a158e5e`, `be5ceb9`; results in `results/explore/`). Each run records its SHA, config, data fingerprint and selection log. 259 CPU tests pass, and every stage reproduces locally (`results/final_audit/REPRODUCTION.md`, `docs/REPRODUCIBILITY.md`).